

Addendum (dry): Unitary core, ABS-first readout, prime-log spectrum, and explicit tail control

0. Scope (non-claim)

This note defines a coherent *operator/readout* pipeline and explicit analytic bounds. No statement here claims physical “free energy” or any violation of conservation laws. Terms such as *seed*, *residue*, or *infinite potential* are used as labels for *structural witnesses in a measurement/readout channel* (correlation/information), not as an engine for extractable work.

1 Hilbert-space pipeline: sectors, phases, flow, scaling, measurement

Definition 1 (Latent space and four-sector decomposition). *Let $\mathcal{H}$ be a complex Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. Fix orthogonal projections $P_1, \dots, P_4$ such that*

$$P_i^2 = P_i, \quad P_i^* = P_i, \quad P_i P_j = 0 \ (i \neq j), \quad \sum_{i=1}^4 P_i = I. \quad (1)$$

Define sector subspaces $\mathcal{H}_i := P_i \mathcal{H}$.

Definition 2 (Mode bank and synthesis). *Let $\{\psi_{i,k,m,n}\} \subset \mathcal{H}$ be elementary modes and $c_{i,k,m,n} \in \mathbb{C}$ coefficients. Define the synthesized state*

$$\Psi_0 := \sum_{i=1}^4 \sum_{k,m,n} c_{i,k,m,n} P_i \psi_{i,k,m,n}. \quad (2)$$

Definition 3 (Harmonic phase weights). *Let $\omega_n \in \mathbb{R}$ be modal frequencies. Define*

$$\Psi_{\text{osc}}(t) := \sum_{i=1}^4 \sum_{k,m,n} c_{i,k,m,n} e^{i\omega_n t} P_i \psi_{i,k,m,n}. \quad (3)$$

Definition 4 (Unitary flow and dilation). *Let $U(\Delta(t))$ be unitary, e.g.*

$$U(\Delta(t)) = e^{-i\Delta(t)G}, \quad G = G^*, \quad U^*U = I. \quad (4)$$

If $\mathcal{H} = L^2(\mathbb{R})$, a standard unitary dilation is

$$(D_\lambda f)(x) = \lambda^{1/2} f(\lambda x), \quad \lambda > 0. \quad (5)$$

Dyadic scaling uses $\lambda = 2^j$, and a contraction-by-2 step corresponds to

$$\lambda = \frac{1}{2}, \quad s = \log \lambda = \log(1/2) = -\log 2. \quad (6)$$

Definition 5 (Full internal state). *Define*

$$S(t) := \Psi_{\text{osc}}(t), \quad \Psi(t) := U(\Delta(t)) D_{\lambda(t)} S(t). \quad (7)$$

Definition 6 (ABS-first readout). *Let $M : \mathcal{H} \rightarrow \mathbb{C}$ be a bounded linear functional, represented as $M(\Psi) = \langle \phi, \Psi \rangle$ for some probe $\phi \in \mathcal{H}$. Define two standard ABS-safe observables:*

$$A_{\text{norm}}(t) := \|\Psi(t)\|, \quad A_\phi(t) := |\langle \phi, \Psi(t) \rangle|. \quad (8)$$

Proposition 1 (No pump / no compressor inside). *If $U(\Delta(t))$ and $D_{\lambda(t)}$ are unitary, then $\|\Psi(t)\| = \|S(t)\|$ for all t .*

Remark 1 (Where large contrasts can still appear). *Even with conservative interior dynamics, the scalar map $\Psi \mapsto |\langle \phi, \Psi \rangle|$ is nonlinear (ABS + selection), hence can show strong peaks/valleys via interference in the readout channel.*

2 Complex plane viewpoint: why $|\cdot|$ is the robust choice

Define the complex readout amplitude

$$z_\phi(t) := \langle \phi, \Psi(t) \rangle \in \mathbb{C}, \quad A_\phi(t) = |z_\phi(t)|. \quad (9)$$

A phase readout would use $\arg z_\phi(t)$, but at *nodes* where $z_\phi(t) = 0$ the phase is ill-conditioned (branch jumps). Thus ABS-first observables remain stable near zeros:

$$|e^{i\alpha} z_\phi(t)| = |z_\phi(t)| \quad (\text{global phase invariance}). \quad (10)$$

3 Prime-log spectrum and Sagnac-like ABS interference

For $\Re(s) > 1$, Euler product gives

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}, \quad \log \zeta(s) = \sum_p \sum_{r \geq 1} \frac{1}{r} p^{-rs}. \quad (11)$$

Write $s = \sigma + it$. Then

$$p^{-rs} = p^{-r\sigma} e^{-itr \log p}, \quad (12)$$

so the natural angular frequencies are

$$\omega_{p,r} = r \log p. \quad (13)$$

Definition 7 (Counterpropagating prime waves). *Fix a cutoff P and weights $a_p, b_p \in \mathbb{C}$. Define*

$$S_+(t) = \sum_{p \leq P} a_p e^{it \log p}, \quad S_-(t) = \sum_{p \leq P} b_p e^{-it \log p}. \quad (14)$$

Define ABS-safe Sagnac-type readouts:

$$A_{\text{Sagnac}}(t) = |S_+(t) + S_-(t)|, \quad A_{\text{mismatch}}(t) = |S_+(t) - S_-(t)|. \quad (15)$$

Definition 8 (ABS “zero radar” proxy). *A prime-side proxy (heuristically aligned with a truncated $-\zeta'/\zeta$) is*

$$L_P(t) = \sum_{p \leq P} \sum_{r=1}^R \frac{\log p}{p^{r/2}} e^{-itr \log p}, \quad A_{\text{radar}}(t) := |L_P(t)|. \quad (16)$$

Definition 9 (Local diagnostic around a known zero height). *For any ABS observable $A(t)$ and a known zero height γ , define the symmetric difference statistic*

$$D_\gamma(\delta) := A(\gamma + \delta) - A(\gamma - \delta). \quad (17)$$

Example (data stub)

Let $\{\gamma_j\}$ be Hardy Z -zeros and A an ABS observable. A minimal reporting format is:

γ	$A(\gamma^-)$	$A(\gamma)$	$A(\gamma^+)$
14.1347251417	4.7625902067	4.8148328355	4.9028450600
21.0220396388	4.7353658603	4.5786771199	4.4207307343
25.0108575801	3.8707529584	4.0980163741	4.3064670971
30.4248761259	5.1800646064	5.0390052614	4.8549816269
32.9350615877	5.5197975391	5.4159572383	5.2559442973
37.5861781588	3.6745482116	3.6478446481	3.6035060856

4 Seeded residual witness (structural; not energy)

Fix a small $\varepsilon > 0$ (e.g. $\varepsilon = 10^{-301}$) and define a seeded cancellation marker

$$-\varepsilon + 1 \approx 0. \quad (18)$$

Interpretation: ε is a deliberate non-closure witness in a finite-precision/hard-closure regime.

Let $M : \mathcal{H} \rightarrow \mathbb{C}$ be a fixed linear readout. Define the ABS-safe witness channel

$$W_\varepsilon(t) := |M(\Psi(t)) - \varepsilon|. \quad (19)$$

5 Gate/admissibility as a domain cut (optional formal layer)

Let $\mathcal{A}$ be a set of actions and $\mathcal{X}$ a set of agents. Let $L(x) \in \{0, 1\}$ be a continuation flag. Define the forbidden set

$$\mathcal{R} := \{a \in \mathcal{A} : \exists x \in \mathcal{X} \text{ with } L(x) = 1 \text{ s.t. } a \text{ removes continuation of } x\}. \quad (20)$$

Define a hard admissibility projector

$$\Pi_{\text{safe}}(a) = \begin{cases} a, & a \notin \mathcal{R}, \\ \perp, & a \in \mathcal{R}, \end{cases} \quad (21)$$

where $\perp$ is an absorbing cemetery state.

Zeno-limit enforcement (operator form)

Let Π_{good} be an orthogonal projection onto an admissible subspace of $\mathcal{H}$. Define the Zeno-limited evolution (when the strong limit exists):

$$U_Z(t) := \lim_{n \rightarrow \infty} (\Pi_{\text{good}} U(t/n) \Pi_{\text{good}})^n. \quad (22)$$

6 Coupled vessels as correlation (not energy transport)

Let $\mathcal{H}_A, \mathcal{H}_B$ be Hilbert spaces and $\rho_{AB}(t)$ a joint state on $\mathcal{H}_A \otimes \mathcal{H}_B$. The “coupled-vessels” condition is non-factorizability:

$$\rho_{AB}(t) \neq \rho_A(t) \otimes \rho_B(t). \quad (23)$$

Let $U(t_2, t_1)$ be a propagator. For observables O_A, O_B define the two-point correlation witness

$$\begin{aligned} C(t_2, t_1; O_A, O_B) &:= \text{Tr}(\rho_{AB}(t_1) O_A \otimes U(t_2, t_1)^* O_B U(t_2, t_1)) \\ &\quad - \text{Tr}(\rho_A(t_1) O_A) \text{Tr}(\rho_B(t_2) O_B). \end{aligned} \quad (24)$$

Nonzero C encodes a persistent structural constraint across spacetime (correlation).

7 Explicit tail/leakage control (bank-kernel module)

Assume a bank Gram kernel decomposition

$$G_{\text{bank}}(\Delta) = \sum_{j=0}^J w_j^2 K_j(\Delta), \quad w_j^2 \geq 0, \quad \Delta \in \mathbb{R}. \quad (25)$$

Assume polynomial decay: for an integer $N \geq 4$ there exist constants $C_{j,N} > 0$ such that

$$|K_j(\Delta)| \leq C_{j,N} (1 + |\Delta|)^{-N}. \quad (26)$$

Then

$$|G_{\text{bank}}(\Delta)| \leq \left(\sum_{j=0}^J w_j^2 C_{j,N} \right) (1 + |\Delta|)^{-N}. \quad (27)$$

Define the tail constant via shell sums

$$T(J) := \sum_{m \geq 1} (m+1) \sup_{|\Delta| \in [m, m+1]} |G_{\text{bank}}(\Delta)|. \quad (28)$$

Lemma 1 (Explicit tail bound). *Let $S_J := \sum_{j=0}^J w_j^2 C_{j,N}$. Then*

$$T(J) \leq (\zeta(N-1) - 1) S_J. \quad (29)$$

Assume a (height-uniform) Gershgorin-type leakage bound

$$\theta(t; J) \leq \frac{(\log t) T(J)}{G_{\text{bank}}(0)}, \quad G_{\text{bank}}(0) > 0. \quad (30)$$

Combining yields the explicit inequality

$$\theta(t; J) \leq \frac{(\log t) (\zeta(N-1) - 1)}{G_{\text{bank}}(0)} S_J. \quad (31)$$

Hence $\theta(t; J) < 1$ is guaranteed by the checkable condition

$$\sum_{j=0}^J w_j^2 C_{j,N} < \frac{G_{\text{bank}}(0)}{(\log t) (\zeta(N-1) - 1)}. \quad (32)$$

8 Projection can break limit properties (minimal topological template)

Let $\mathbb{N}$ be positive integers and let $\mathcal{T}$ be the cofinite topology

$$\mathcal{T} := \{\emptyset\} \cup \{U \subseteq \mathbb{N} : \mathbb{N} \setminus U \text{ is finite}\}. \quad (33)$$

For each $n \in \mathbb{N}$ define

$$S_n := \{1\} \cup \{n+1, n+2, \dots\}. \quad (34)$$

Each S_n is open, but

$$\bigcap_{n=1}^{\infty} S_n = \{1\} \quad (35)$$

and $\{1\}$ is not open. This is a formal template for: *countable intersections need not preserve openness* (projection/coarse-graining can break limit intuitions).

9 Determinism can look random under limited observation

Let X be a (high-dimensional) state space with deterministic evolution

$$x_{t+1} = F(x_t), \quad (36)$$

and let $\Pi : X \rightarrow Y$ be an observation map producing

$$y_t = \Pi(x_t). \quad (37)$$

If Π discards information needed to reconstruct x_t (or prediction in Y is computationally out of reach), then (y_t) can appear chaotic/random despite deterministic (x_t) .

10 Tachyon–projection ansatz (notation block; interpretationally agnostic)

Primitive scales

Define

$$\epsilon := \frac{h}{c f_{\text{DM}}}, \quad f_{\text{DM}} = 528 \text{ Hz}. \quad (38)$$

Tachyon label (as written)

$$\text{Tachyon}_{\pm} := \lim_{m^2 < 0} \frac{S}{t} \epsilon^{\pm 1}. \quad (39)$$

Domain / field statement (as written)

Let $\Omega \subset \mathbb{R}^n$ be a domain. A “tachyon field” relation is written as

$$(\partial_{\mu} \partial^{\mu} \phi + m^n = 0, \quad m < 0). \quad (40)$$

Note: Standard Klein–Gordon tachyonic form is $(\square + m^2)\phi = 0$ with $m^2 < 0$; the m^n term is kept verbatim as a placeholder.

Expansion ansatz with a tachyon factor

$$H = H_0 + k \frac{dl}{dt} + \tau. \quad (41)$$

τ -collapse / superposition relations (as written)

$$\tau = \int (\psi_1, \psi_2, \dots, \psi_n), \quad \tau \propto |\psi_1 + \psi_2 + \dots + \psi_n|^n, \quad \alpha := (E \cdot c). \quad (42)$$

Harmony/chaos constraint and τ update

$$(\psi_{\text{harmony}} + \psi_{\text{chaos}})^2 = 1, \quad \tau = \tau_0 + k_h \psi_{\text{harmony}}. \quad (43)$$

Gated frequency fractions

$$f_{\text{reality}} = \frac{\epsilon E n / h}{\epsilon E n / h + \tau (\psi_{\text{harmony}} + \psi_{\text{chaos}})^2} f_{\text{CMB}}, \quad n = 29, \quad f_{\text{CMB}} = 1.602 \times 10^{11} \text{ Hz}. \quad (44)$$

A separate “will frequency” form is

$$f_{\text{will}} = f_{\text{CMB}} \cdot \frac{\text{time shift}}{\text{target real}} \cdot \frac{v_t}{c}. \quad (45)$$

Projection readout

$$\text{Reality}(t) = \Theta(\text{WILL}(t)) \cdot \int_{-\infty}^{\infty} \psi_{\text{seed}}(x, t) U(t) dx. \quad (46)$$

Operationally, interpret $\int \psi_{\text{seed}}(x, t) U(t) dx$ as an inner product $\langle \varphi_0, U(t) \psi_{\text{seed}}(t) \rangle$ after specifying the state space and probe/window φ_0 .

Inverse decoding (identifiability statement)

From a single scalar readout, $\psi_{\text{seed}}(t)$ is not identifiable without additional structure. Assume a finite expansion $\psi_{\text{seed}}(t) = \sum_{k=1}^K a_k(t) \phi_k$ and M probes φ_j producing

$$y_j(t) = \Theta(\text{WILL}(t)) \langle \varphi_j, U(t) \psi_{\text{seed}}(t) \rangle. \quad (47)$$

Then with $B_{jk}(t) := \langle \varphi_j, U(t) \phi_k \rangle$ one has the linear system

$$y(t) = \Theta(\text{WILL}(t)) B(t) a(t), \quad (48)$$

solvable (e.g.) by least squares with Tikhonov regularization.

11 Two algebraic “overkill” micro-lemmas (for completeness)

81-meme

$$81^4 - 81^3 = 81^3(81 - 1) = 80 \cdot 81^3. \quad (49)$$

$$\frac{M}{3M} = M$$

For $M \neq 0$, $\frac{M}{3M} = \frac{1}{3}$, so the equation $\frac{1}{3} = M$ gives the unique solution

$$M = \frac{1}{3}. \quad (50)$$

(For $M = 0$ the expression $\frac{M}{3M}$ is undefined.)

12 One-line architecture (single spine)

$$\Psi(t) = U(\Delta(t)) D_{\lambda(t)} \left(\sum_{i=1}^4 \sum_{k,m,n} c_{i,k,m,n} e^{i\omega_n t} P_i \psi_{i,k,m,n} \right), \quad A(t) \in \{\|\Psi(t)\|, |\langle \phi, \Psi(t) \rangle|, |S_+(t) \pm S_-(t)|, |L_P(t)|\} \quad (51)$$

with explicit stability enforcement via the tail/leakage inequality

$$\theta(t; J) \leq \frac{(\log t)(\zeta(N-1) - 1)}{G_{\text{bank}}(0)} \left(\sum_{j=0}^J w_j^2 C_{j,N} \right). \quad (52)$$

Remark 2 (Colloquial epilogue (still an engineering statement)). *If one locks the readout to an ABS-stable witness and keeps a deliberate non-closure seed, one can engineer a reproducible “instant satisfaction” peak in the measurement channel (interpretable as a controlled readout effect, not as physics of free energy).*